Dichotomies for #CSP on Graphs That Forbid a Clique as a Minor

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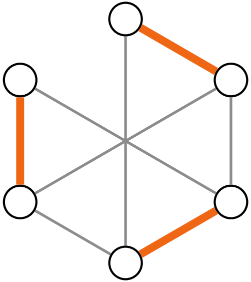
Counting Perfect matchings



$$\#\mathrm{PM}(G) = \sum_{M} \prod_{e \in M} w_e$$

planar $\Longrightarrow$ **P**

general $\Longrightarrow$ **#P-hard**

(bipartite #PM = permanent)

Kasteleyn (1961); Temperley–Fisher (1961); Valiant (1979).

How far beyond planarity?

> A minor-closed class is described
> by finitely many forbidden minors.

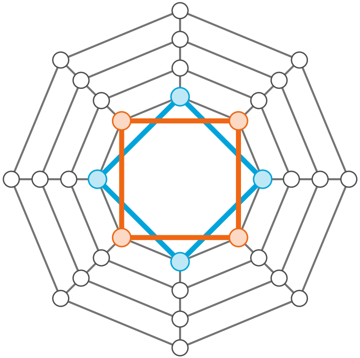
planar $\iff$ forbids K_5 and $K_{3,3}$ as minors

Which forbidden minor makes $\#\mathrm{PM}$ easy?

planar FKT	$K_{3,3}$-free	bounded genus	K_5-free	dichotomy
1961	1989	1999	2014	2022

Wagner (1937); Vazirani (1989); Galluccio–Loebl and Tesler (1999–2000); Straub–Thierauf–Wagner (2014); Curticapean (2014); Thilikos–Wiederrecht (2022).

The obstruction: Shallow Vortex



The shallow vortex H_4

$\mathcal{H}$ = all minors of $H_1, H_2, \ldots$

$fb(\mathcal{G})$ forbids
some graph in $\mathcal{H}$?

yes $\Rightarrow$ **P**

no $\Rightarrow$ **#P-hard**

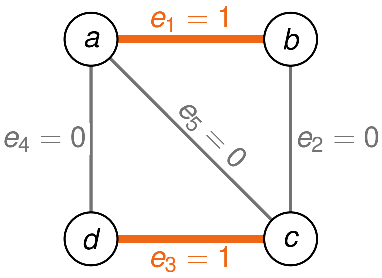
"killing a vortex"

Thilikos–Wiederrecht, "Killing a Vortex" (FOCS 2022).

Hardness mechanism: Curticapean–Xia (2022).

#PM as a Holant instance

$\#\mathrm{PM}$ **vs.** Holant(EXACTONE)



$M^\star = \{e_1, e_3\}$

$\#\mathrm{PM}(G) = \left|\{M^\star, \{e_2, e_4\}\}\right| = 2$

$$f_v = \mathrm{EXACTONE}_{\deg(v)},$$

$$f_v(\mathbf{x}) = \begin{cases} 1, & \text{exactly one input is } 1, \\ 0, & \text{otherwise.} \end{cases}$$

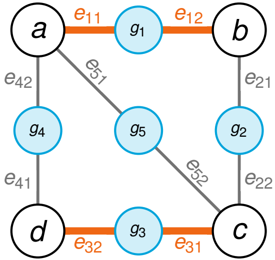
Let $\sigma_i = \sigma(e_i)$.

$$\begin{aligned} \mathrm{Holant}_G = & \sum_{\sigma \in \{0,1\}^5} f_a(\sigma_1, \sigma_4, \sigma_5) f_b(\sigma_1, \sigma_2) \\ & \cdot f_c(\sigma_2, \sigma_3, \sigma_5) f_d(\sigma_3, \sigma_4) \\ = & \ 2. \end{aligned}$$

$$\sigma \in \{(1,0,1,0,0), (0,1,0,1,0)\}.$$

Edge weights as binary signatures

Holant(EXACTONE, w)



$$w(M) = \prod_{e \in M} w(e)$$

$$\#\mathrm{PM}_w(G) = \sum_{M \in \mathcal{M}(G)} w(M)$$

$$= w_1 w_3 + w_2 w_4$$

For e_i, let $w_i = w(e_i)$.

$$g_{e_i}(x, y) = \begin{cases} 1, & (x, y) = (0, 0), \\ w_i, & (x, y) = (1, 1), \\ 0, & x \neq y. \end{cases}$$

$$\mathrm{Holant}_G = \sum_{\sigma \in \{0,1\}^{10}} \prod_{i=1}^{5} g_{e_i}(\sigma_{i1}, \sigma_{i2})$$

$$\cdot f_a(\sigma_{11}, \sigma_{42}, \sigma_{51}) f_b(\sigma_{12}, \sigma_{21})$$

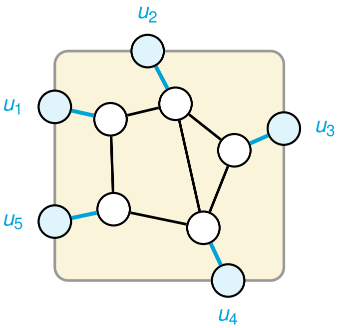
$$\cdot f_c(\sigma_{22}, \sigma_{31}, \sigma_{52}) f_d(\sigma_{32}, \sigma_{41})$$

$$= w_1 w_3 + w_2 w_4 = \#\mathrm{PM}_w(G).$$

Matchgates and permutability

Matchgates $\mathscr{M}$

$$f_\Gamma(\alpha) = \#\mathrm{PM}(\Gamma - X_\alpha)$$



Permutable matchgates $\mathscr{M}_P$

permutable

$\Downarrow$

remains a matchgate under every variable permutation

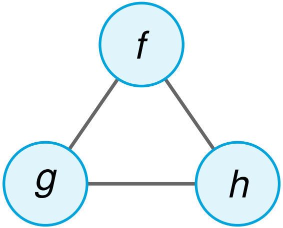
$\Downarrow$

does not depend on the embedding

Valiant's matchgate framework; Cai–Choudhary (2007); Meng (2025) for permutable matchgates.

PMG Holant to #PM

Holant instance G

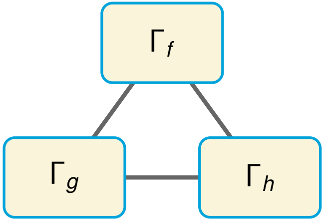


$f, g, h \in \mathscr{M}_P$

replace each signature

$f_v \mapsto \Gamma_{f_v}$

Weighted graph $G^\star$



matchgate expansion

$$\boxed{Z(G) = \#\mathrm{PM}(G^\star)}$$

G planar $\Longrightarrow$ $G^\star$ planar $\xRightarrow{\text{FKT}}$ **P**.

How far does PMG survive?

local constraint
$f_v \in \mathscr{M}_P$

host graph
$G \in fb(\mathcal{G})$

$$\text{classify}\,(\mathcal{F},\ fb(\mathcal{G}))$$

bounded degree: kill a shallow vortex

unbounded degree: also kill an apex

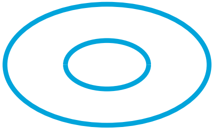
Genus is always manageable

Clique-Minor Boundaries for PMG

Forbidden minor(s)	$K_5, K_{3,3}$	K_5	K_6	K_7	K_8
Unbounded degree	**P**	**P**	**#P-hard**	**#P-hard**	**#P-hard**
Bounded degree	**P**	**P**	**P**	**P**	**#P-hard**

Three features, three roles

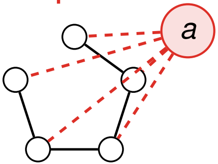
genus



bounded genus
$\Rightarrow$ FKT-like alg

manageable

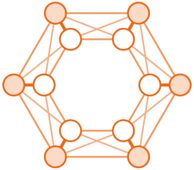
apex



degree-dependent
bounded: easy
unbounded: hard

degree sensitive

vortex



introduce one $\Rightarrow$
#P-hardness

hard

VGA-decomposition framework

VGA decomposition of width k

$=$ tree decomposition $+$ each torso G_t:

1. $\text{size}(G_t) \leq k$; or
2. G_t is embeddable on a surface
   of genus $\leq k$
   with $\leq k$ apices
   and $\leq k$ vortices.

VGA-decomposition framework

-GA decomposition of width k

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VGA-decomposition framework

-G- decomposition of width k

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1. size$(G_t) \leq k$; or

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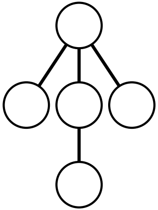
K_5 easy, K_6 hard for unbounded degree

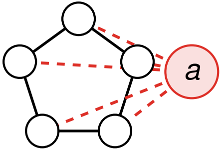
K_5-minor-free



-G- decom-
position
of width k

compress from leaves $\Rightarrow$ **P**

K_6-minor-free

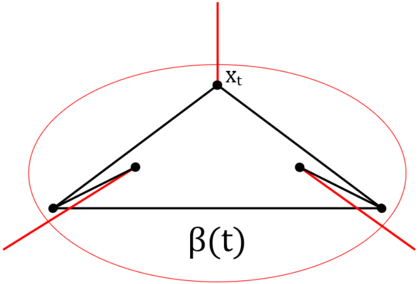


contains
every apex
graph

one apex $\Rightarrow$ **#P-hard**

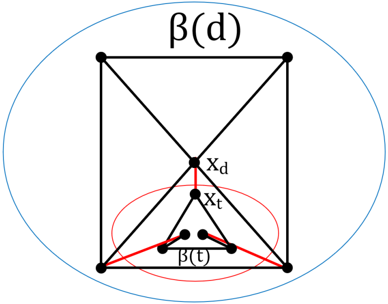
Signature realizations are omitted from this talk.

K_5: compress a subtree



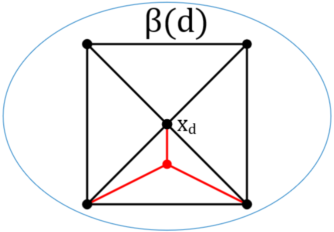
split the child and parent sides by path gadgets

K_5: compress a subtree



evaluate the child side through at most three boundary edges

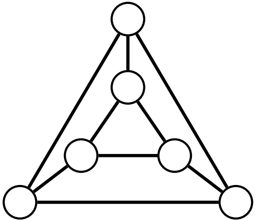
K_5: compress a subtree



replace the whole child subtree by one PMG vertex

parity $+$ arity $\leq$ 3 keeps the new signature inside $\mathscr{M}_P$

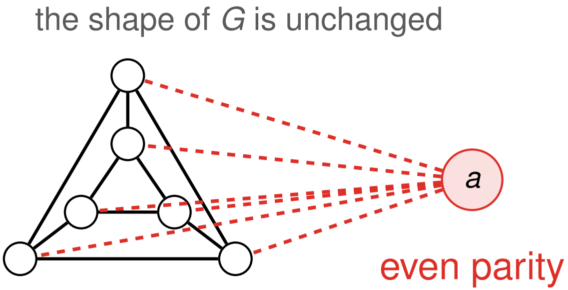
K_6: one apex is hard



planar cubic graph G

Hard starting problem: Xia–Zhang–Zhao (2007); planar pairing realization: Guo–Williams (2020).

K_6: one apex is hard



$$Z(G_a) = \#\{\text{matchings of } G\}$$

Hard starting problem: Xia–Zhang–Zhao (2007); planar pairing realization: Guo–Williams (2020).

A Hadamard transform returns to #CSP

Boolean $\#\mathrm{CSP}$ incidence graph	$\equiv$	Holant$(\mathcal{F} \mid \mathcal{EQ})$	$\xrightarrow{H_2}$	Holant$(\widehat{\mathcal{F}} \mid \widehat{\mathcal{EQ}})$ PMG regime

$\mathcal{F} \subseteq \mathscr{A}$ or $\mathscr{P}$	tractable on every graph class
$\widehat{\mathcal{F}} \subseteq \mathscr{M}_P$	use the minor boundary
otherwise	hard already on planar graphs

Holographic transformation: Valiant; planar Boolean $\#\mathrm{CSP}$ dichotomy: Cai–Fu (2017).

PMG signatures + minor structure + degree

genus is manageable

an apex is the **unbounded-degree** boundary

a vortex is the **bounded-degree** boundary

Thank you!

Questions and comments are welcome.